Knowledge Management:

An Introduction and Perspective

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Leaders of successful organizations are consistently searching for better ways to improve performance and results. Frequent disappointments with past management initiatives have motivated managers to gain new understandings into the underlying, but complex mechanisms – such as knowledge – which govern an enterprise's effectiveness. Knowledge Management, far from being a management 'fad', is broad, multi-dimensional and covers most aspects of the enterprise's activities. To be competitive and successful, experience shows that enterprises must create and sustain a balanced intellectual capital portfolio. They need to set broad priorities and integrate the goals of managing intellectual capital and the corresponding effective knowledge processes. This requires systematic Knowledge Management. With knowledge as the major driving force behind the 'economics of ideas', we can expect that the emphasis on knowledge creation, development, organization and leverage will continue to be the prime focus for improving society.

INTRODUCTION

In the mid-1980s, individuals and organizations began to appreciate the increasingly important role of knowledge in the emerging competitive environment. International competition was changing to increasingly emphasize product and service quality, responsiveness, diversity and customization. Some organizations, such as US-based Chaparral Steel, had been pursuing a knowledge focus for some years, but during this period it started to become a more wide-spread business concern. These notions appeared in many places throughout the world – almost simultaneously in the way bubbles appear in a kettle of superheated water! Over a brief period from 1986 to 1989, numerous reports appeared in the public domain concerning how to manage knowledge explicitly. There were studies, results of corporate efforts, and conferences on the topic.

In spite of the wide geographical distribution, most professional managers did not realize the importance of explicit and systematic Knowledge Management (KM) – and this realization is still limited. The observations by Quinn, Anderson and Finkelstein in the *Harvard Business Review* that "Surprisingly little attention has been given to managing professional intellect" and by Suchman in the *Communications of the ACM* that "How people work (with their minds) is one of the best kept secrets in America" may be all too valid.

In a 1989 survey, several *Fortune* 50 CEOs agreed that knowledge is a fundamental factor behind an enterprise's success and all its activities.[1] They opined that enterprise viability hinges directly upon the competitive quality of the knowledge assets and their successful exploitation. Leaders of progressive organizations and nations are pursuing ways to create and generate value from knowledge assets within their organizations. Often, personal beliefs spur these efforts, paired with strong convictions that competitive knowledge assets and their effective utilization are critical for success. Less frequently, we find careful analyses and well-founded theories. The explicit focus on knowledge is so recent that business practitioners still lead KM exploration and implementation work. There is limited support from academic and management research, except in specialized technical areas such as applied artificial intelligence and use of information technology.

No general approach to managing knowledge is commonly accepted, although several isolated, and at times diverging, notions are being advanced. One notion deals with management of explicit

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Leveraging Human Capital

Marc Baker, Mike Barker, Jon Thorne and Martin Dutnell

Summary

This paper introduces some of the elements of Knowledge Management and outlines the approaches RM Consulting – the internal consultancy of the Royal Mail and the Post Office – is taking to support the development of knowledge enablers within the organization. A Knowledge Management approach is described which focuses on people (including the organizational structure), processes and technology. The initial Knowledge Management focus of the organization is aimed principally on explicit knowledge, and the establishment of tools which help to capture internal information. The long-term challenge is to capture the knowledge of the organization in such a way that all of its employees can maximize the value they provide to the organization's stakeholders.

Keywords

Explicit knowledge
Information technology
Knowledge Management
Tacit knowledge
RM Consulting
Virtual teams

Knowledge Management: An Emerging Discipline

Syed Z. Shariq

Summary

Society is entering into an era where the future essentially will be determined by people's ability to wisely use knowledge, a precious global resource that is the embodiment of human intellectual capital and technology. As people begin to expand their understanding of knowledge as an essential asset, they are realizing that in many ways the future is limited only by imagination and the ability to leverage the human mind. As knowledge increasingly becomes the key strategic resource of the future the need to develop comprehensive understanding of knowledge processes for the creation, transfer and deployment of this unique asset are becoming critical. Educational institutions and training organizations and businesses and knowledge-based organizations in the public sector are in need of an integrative discipline for studying, researching and learning about the knowledge assets – human intellectual capital and technology. An international society of knowledge professionals is proposed which can provide the necessary focus for fostering collaboration among the best minds and organizations on study, research and learning dedicated to the underlying disciplines and their integrative evolution into the emergence of Knowledge Management as a new discipline.

Keywords

Intellectual capital
Knowledge economy
Knowledge Management
Knowledge processes
Knowledge professionals
Knowledge-based organizations

Knowledge Management Benchmarks

Rory L. Chase

Summary

Knowledge Management is now one of the major driving forces of organizational change and wealth creation. This paper reviews some of the major concepts and approaches as discussed at a recent international congress on the subject. Beginning with an examination of some of the factors propelling the global knowledge economy, the paper then explores knowledge-based organizational strategy, illustrated by a number of case studies from leading practitioners, including British Petroleum, Glaxo Wellcome, ICL, Nokia Telecommunications, the UK Post Office and Zeneca Pharmaceuticals. The concept of intellectual capital lies at the heart of Knowledge Management. Some companies define intellectual capital in terms of value creation, for others it is value extraction. The two different approaches, illustrated by Skandia and the Dow Chemical Company, are reviewed, along with a new tool for measuring intellectual capital.

Keywords

Culture change
Innovation
Intellectual capital
Knowledge creation
Knowledge economy
Knowledge Management

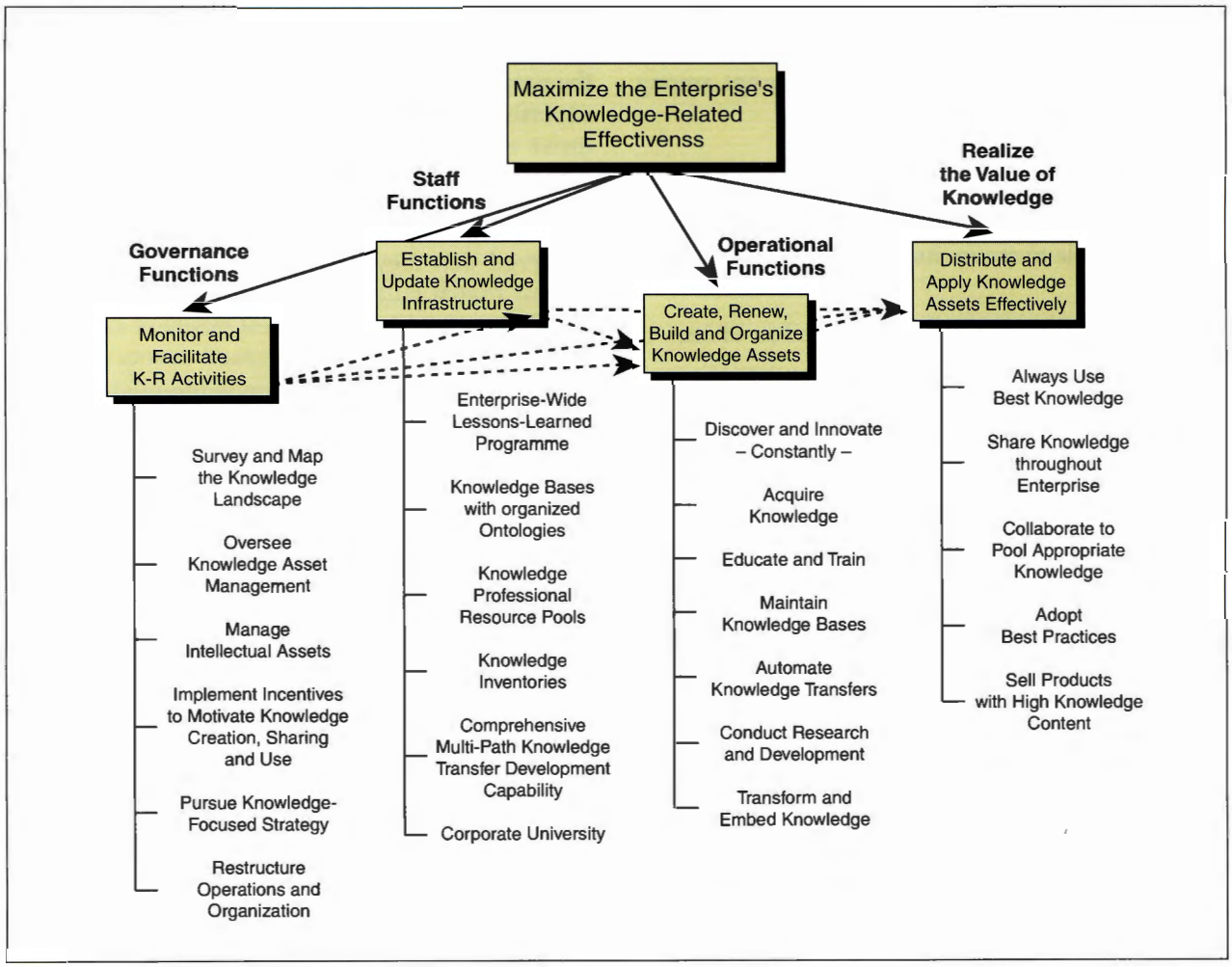


Figure 1: Four areas of Knowledge Management emphasis

knowledge using technical approaches. It primarily focuses on knowledge acquired from people, in computer knowledge bases, knowledge-based systems, and knowledge made available over technology-based networks using e-mail, groupware, and other tools.[2] A second notion focuses on management of 'intellectual capital' in the forms of structural capital and human capital in people.[3] A third notion for managing knowledge has a broader focus to include all relevant knowledge-related aspects which affect the enterprise's viability and success. It encompasses the above notions to also include most other knowledge-related practices and activities of the enterprise.[4]

KNOWLEDGE MANAGEMENT – A WORKING DEFINITION

KM has emerged only recently as an explicit area of pursuit in managing organizations – and even more recently as a topic of serious study or academic knowledge transfer. Knowledge and expertise clearly have been managed implicitly as long as work has been performed. The first hunters surely were concerned about the expertise and skills of their team mates when they went out to capture prey. They also, we must surmise, ascertained that what they knew as the best and most successful practices were taught to up-and-coming hunters to ensure the long-term viability of the group. From very early times, wise people have secured sustained succession by transferring in-depth knowledge to the next generation.

Faculties within universities and other learning institutions have been concerned about knowledge transfer processes and the creation and application of knowledge for several millennia. Early on, Indian mathematicians built upon generations of knowledge to develop mathematics that is quite sophisticated even by today's standards. Phoenicians were implicitly concerned about how knowledge about trade logistics and merchant practices was built, transferred to employees and applied to make operations as successful as possible. Nevertheless, systematic KM for business

purposes, as we understand it today, did not become explicit until about a decade ago and even today it is not a commonly shared concept among managers.

Simply stated, the objectives of KM are:

1. To make the enterprise act as intelligently as possible to secure its viability and overall success.
2. To otherwise realize the best value of its knowledge assets.

To reach these goals, advanced organizations build, transform, organize, deploy and use knowledge assets effectively. Stated differently, the overall purpose of KM is to maximize the enterprise's knowledge-related effectiveness and returns from its knowledge assets and to renew them constantly. KM is to understand, focus on, and manage systematic, explicit, and deliberate knowledge building, renewal, and application – that is, manage effective knowledge processes (EKP).

From a managerial perspective systematic KM comprises four areas of emphasis:

1. Top-down monitoring and facilitation of knowledge-related activities.
2. Creation and maintenance of the knowledge infrastructure.
3. Renewing, organizing, and transforming knowledge assets.
4. Leveraging (using) knowledge assets to realize their value.

These areas are shown in Figure 1 which also indicates some relevant knowledge-related practices and activities.[5]

ECONOMICS AND STRATEGY GUIDE THE WAY

The 'economics of ideas' describes the almost unlimited potential for economic growth and success that new innovations and knowledge-based products make possible.[6] This diverges from more traditional economic perspectives which presume restricted expansion opportunities based on scarcity of physical resources, available labour, capital, etc. In contrast to earlier theories, the 'economics of ideas' explains much about the increased quality of life and wealth creation of recent decades. Making people knowledgeable brings innovation and continued ability to create and deliver products and services of the highest quality. It also requires effective knowledge capture, reuse, and building upon prior knowledge.

Individual organizations focus on different areas to conduct their business. This reflects their strengths, the nature of their business and the inclinations and expertise of their personnel. In practice, we find that most enterprises pursue one or more of the following Knowledge Management strategies.

In our work with many organizations, we have observed that they pursue different KM strategies to best match culture, priorities, and capabilities. They attempt to derive the best business value from their existing knowledge-based assets or try to create new, competitive knowledge-related assets where that is required. To achieve that, we see that they tend to pursue one or several of five basic knowledge-centred strategies:

1. *Knowledge strategy as business strategy* – a focus on knowledge creation, capture, organization, renewal, sharing and use to have the best possible knowledge available – and used – at each point of action.
2. *Intellectual asset management strategy* – a focus on knowledge enterprise-level management of specific intellectual assets such as patents, technologies, operational and management practices, customer relations, organizational arrangements and other structural knowledge assets.
3. *Personal knowledge asset responsibility strategy* – a focus on personal knowledge responsibility for knowledge-related investments, innovations and the competitive state, renewal, effective use, and availability to others of the knowledge assets within each employee's area of accountability to being able to apply the most competitive knowledge to the enterprise's work.
4. *Knowledge creation strategy* – a focus on knowledge learning, basic and applied research and development, and motivation of employees to innovate and capture lessons learned to obtain new and better knowledge that will lead to improved competitiveness.
5. *Knowledge transfer strategy* – a focus on knowledge systematic approaches to transfer – obtain, organize, restructure, warehouse or memorize, repackage for deployment and distribute – knowledge to points of action where it will be used to perform work. Includes knowledge sharing and adopting best practices.

The fundamental strategies pursued by leading enterprises provide important perspectives on the

driving forces behind the evolution from early agrarian societies to today's knowledge-dependent situation. Organizations focus on different areas to conduct business. In their strategies they reflect their strengths, nature of their business, inclinations and expertise of their personnel, and particularly their fundamental beliefs of what is required to succeed.

The value discipline model advanced by Treacy and Wiersema is particularly helpful in understanding recent changes in strategy focus from the industrial revolution to the knowledge society. Their model posits that highly successful enterprises have value disciplines that focus on either operational excellence, product leadership or customer intimacy. They suggest that only outstanding organizations can pursue more than one of these value disciplines which they define as:[7]

- *Operational excellence* – emphasize leadership in price and customer convenience by minimizing overhead costs, eliminating intermediate production steps, reducing transaction and 'friction' costs and optimizing business processes.
- *Product leadership* – emphasize creation of a stream of state-of-the-art products and services by being creative, commercializing ideas quickly and relentlessly pursuing new solutions, often by obsolescing their own products.
- *Customer intimacy* – emphasize tailoring and shaping products and services to fit an increasingly better definition of the customer's needs to personalize offerings to make the customer successful.

The choice of which KM strategy to pursue is typically based on other strategic thrusts and the value discipline that the enterprise pursues, challenges it faces, and opportunities it wishes to act upon.

AN EVOLUTIONARY PERSPECTIVE OF KNOWLEDGE MANAGEMENT

Treacy and Wiersema's model illustrates the increased reliance on knowledge when observing how the economic focus has shifted over time. Previously, the focus was on making the most with limited resources (pursuing operational excellence). Later, the focus shifted to making clever products (pursuing product leadership). Recently, advanced organizations focus on creating ingenious solutions and developing broad relationships to make customers succeed in their own business (pursuing customer intimacy).

Arguably, the present emphasis on Knowledge Management has resulted naturally from the economic, industrial, and cultural developments which have taken place. In the opinion of many management pundits, we have already entered into the 'knowledge society'. This notion is based on the emphasis on adding competitive value to products and services by application of direct or embedded human expertise – knowledge. This is a considerable change from providing value by relying on natural resources or operational efficiency as was the case in previous eras.[8]

Given this framework, we can construct a perspective of the evolution that has led to today's importance of KM. Historic developments may be portrayed by the following stages of dominant economic activities and foci:

- *Agrarian economies* – creating products for consumption and exchange.
- *Natural resource economies* – natural resource exploitation dominate while customer intimacy was pursued separately by expert tradesmen and guilds.
- *Industrial revolution* – operational excellence through efficiency.
- *Product revolution* – product leadership through variability and sophistication.
- *Information revolution* – continued focus on operational excellence and product leadership.
- *Knowledge revolution* – new focus customer intimacy.

The need to focus on managing knowledge within the enterprise results from both economic and market-driven requirements created by customer demands and international competition. During recent decades, customers have become increasingly discriminating. They demand products and services that fulfil their particular needs more precisely and to greater advantage. It often is not enough to provide generic or commodity products – however sophisticated. Customers – individual consumers and industrial companies alike – require products and services that will make them more successful in their own pursuits and provide them with the best possible advantages.

At the same time as customer sophistication has increased, education and skills of craftspeople and professionals all over the world have risen. Many

former 'developing nations' now can compete with advanced industrial nations in engineering, computer software creation, advanced consumer product design and expert customer service for high-technology products, just to name a few. Competition for who can provide the best products and services based on relevant knowledge has truly become international.

Advanced companies in the Americas and Europe are well aware of the needs to manage knowledge – and to do it systematically and comprehensively. However, the present KM practices are only emerging at this time and most practitioners, of necessity, focus on relatively narrow application areas. This is expected to continue until the availability of good KM methods and practices becomes more common. Nevertheless, companies which have pursued KM for some time are able to point to considerable tangible and intangible benefits. The advantages of deliberate KM are also made apparent by their leadership positions within their industries and markets.

Explicit and systematic management of knowledge has emerged naturally as a result of several developments. After World War II, socio-economic and business environments led to changes in the demand for knowledge-based products and services. In the late 1950s, the emergence of information technology (IT) led to the first steps in automating intelligent behaviour by artificial intelligence (AI) – for research and also for economic gains. In the 1960s, our understanding of business operations, in the forms of operations research (OR) and management sciences, strategic planning and applied cybernetics and 'systems thinking', became better established. This allowed us to think of 'business processes' and their interactions, internal operations and dynamic characteristics in ways not done earlier. Our understanding of how people think and reason has also gradually improved over the years, but was brought forward by cognitive sciences work beginning in the 1970s. Lastly, our understanding of knowledge-based organizational behaviours such as individual and group decision-making were elucidated in the 1980s.[9]

A PARTIAL KNOWLEDGE MANAGEMENT TIME-LINE

All of these developments came together to enable us to consider how knowledge might be managed effectively and systematically. The foundation for 'Knowledge Management' was established and emerged in many organizations in different disguises. Accordingly, some examples of KM-related developments which have taken place over the last two decades are listed below:[10]

1975 – As one of the first to adopt knowledge-focused management, Chaparral Steel bases its internal organizational structure and corporate strategy to rely directly on explicit management of knowledge.

1980 – Digital Equipment Corporation (DEC) installs the first large-scale knowledge-based system (XCON).

1981 – Arthur D. Little starts the Applied Artificial Intelligence Center.

1986 – The concept of 'Management of Knowledge: Perspectives of a New Opportunity' is introduced in a keynote address at a European management conference.

1987 – The first KM book is published in Europe (Sveiby and Lloyd). The first roundtable KM conference *Knowledge Assets into the 21st Century* is hosted by DEC and the Technology Transfer Society at Purdue University.

1989 – The *Sloan Management Review* publishes its first KM-related article (Stata). Several management consulting firms start internal and external efforts to manage knowledge. The International Knowledge Management Network is started in Europe. A survey of *Fortune* 50 CEOs' perspectives on KM by Wiig is undertaken

1990 – The Initiative for Managing Knowledge Assets (IMKA) commences. The first books on the learning organization are published in Europe and the US by Garratt, Senge and Savage.

1991 – Skandia Insurance creates the position of Director of Intellectual Capital. The first Japanese book relating to KM is published in the US (Sakaiya). *Fortune* runs the first article on KM (Stewart). *Harvard Business Review* runs its first article on KM (Nonaka).

1992 – Steelcase and EDS co-sponsor a conference on Knowledge Productivity.

1993 – In Europe, an important KM article is published on "Corporate Knowledge Management" (Steels). The first book explicitly dedicated to KM is published (Wiig).

1994 – Several large consulting firms offer KM services and start seminars for prospective clients on KM. The International Knowledge Management Network expands its scope to include the Internet; publishes a KM survey of 80 Dutch companies

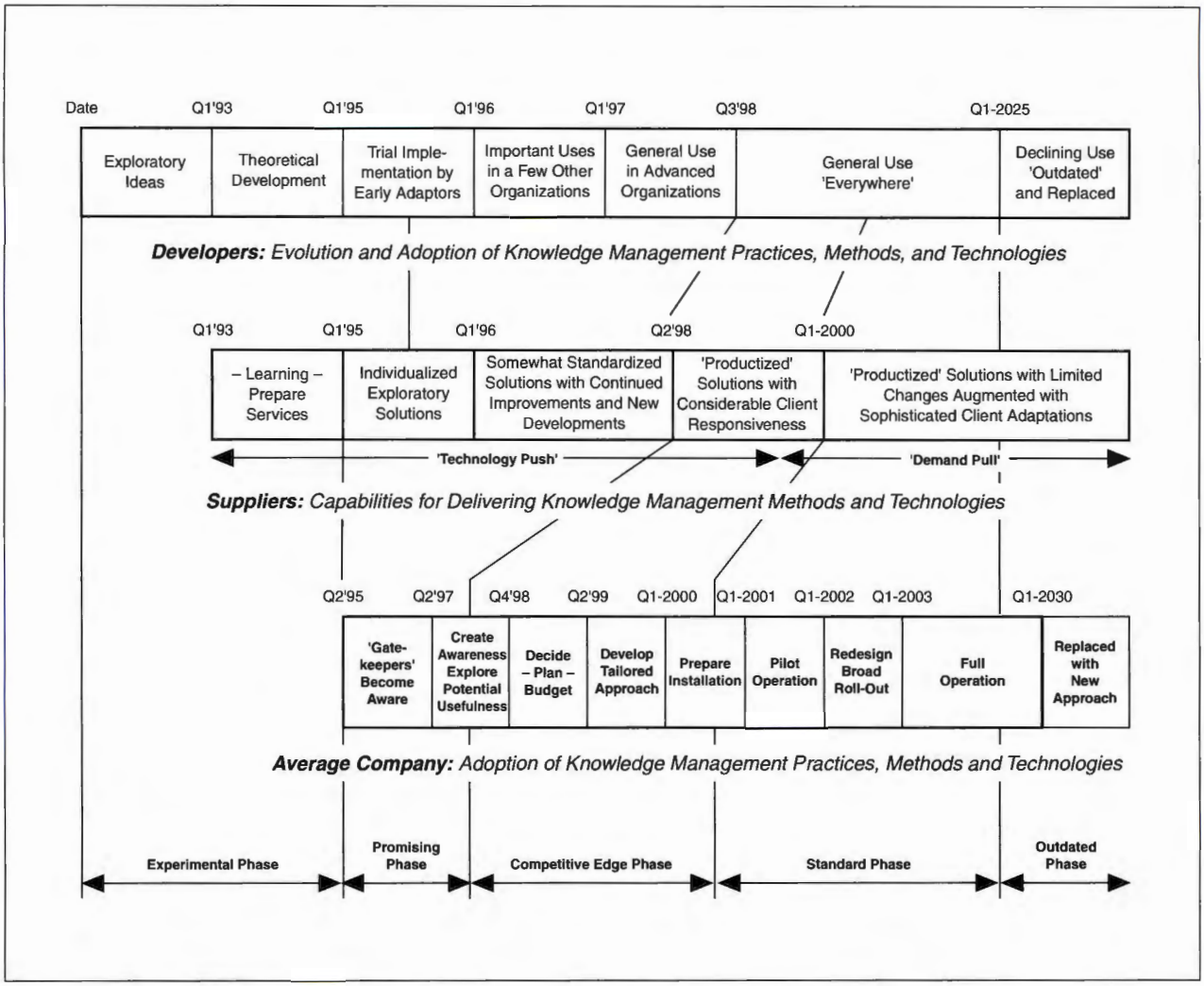


Figure 2: Possible scenario for adopting KM methods, practices and technologies

(Spijkervet and van der Spek); and conducts a conference *Knowledge Management for Executives*. Université de Technologie de Compiègne (France) holds its first annual KM conference. Knowledge Management Network and *FAST Company* magazine are founded in the US.

1995 – The European ESPRIT programme includes explicit requests for KM-related projects. American Productivity & Quality Center (APQC) and Arthur Andersen conduct the *Knowledge Imperatives Symposium* with over 300 attendees. Other KM conferences and seminars are held in the US and Europe. APQC initiates a multi-client KM Consortium Benchmarking Study with 20 sponsors. The Knowledge Management Forum is started on the Internet. A few 'Chief Knowledge Officers' (or equivalent) are appointed.

1996 – Several KM conferences and seminars are held in Europe and the US – organized by both general conference organizers and consulting organizations. Over one dozen large consulting organizations and many smaller ones offer KM services to clients. Many companies are starting KM efforts – some with internal resources only, others with assistance by external organizations. The European Knowledge Management Association is started. The publication *Knowledge Inc.* is started.[11] Many organizations appoint executives responsible for managing knowledge.

1997 – Numerous KM conferences are held in the US, Europe, Asia, Africa; several KM journals are started and many case histories of successful KM efforts and practices are reported. The European Union organizes a KM conference. KM topics are frequent topics in management journals and multiple KM-related books are published.[12] Many more organizations appoint KM executives.

Given these events, it seems clear that KM has attained a considerable – but still narrow – momentum. Many organizations either have not yet heard about KM or have decided to wait. Of a number of CEOs that we contacted, only half of them had heard of KM and few of the CEOs or their

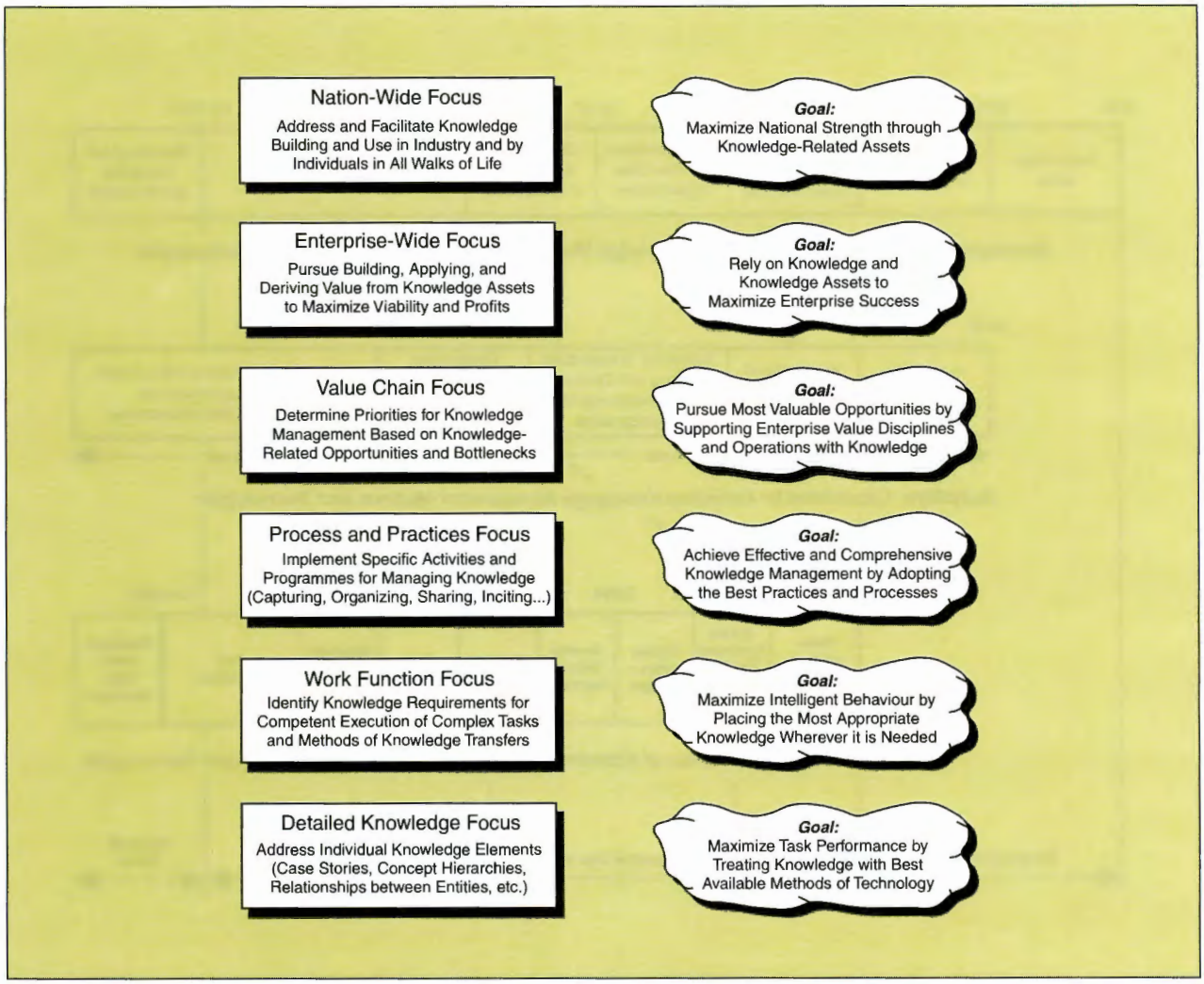


Figure 3: KM focus and goals at different societal and enterprise levels

assistants had any understanding of managing knowledge.

As indicated in Figure 2, we can expect that KM methods and technologies generally will be provided in a 'technology push' approach for some time to come, perhaps until after the turn of the century. Later, we can expect user organizations to seek KM products and services in a 'demand pull' fashion. That will lead to a well-established development and supply chain with accepted and tested methods and tools.

A BROADER PERSPECTIVE

The Knowledge Management focus varies considerably depending upon which societal or enterprise level is involved. On national and local levels, concerns are broad with systematic focus on how knowledge should be managed to benefit the viability of the societal unit. Some relevant foci and goals are indicated in Figure 3.

Examples are the emphasis on knowledge in Singapore, the Netherlands, the Scandinavian countries, the inclusion of KM perspectives in ESPRIT projects, and the recent establishment of a knowledge focus by OECD. These developments are crucial to the future strength and international competitive position of the societies involved. The expressed concerns cover a wide range. One issue is the quality and scope of industrial and university-level educational programmes. Another deals with which concepts and meta-knowledge children need in their first school years to secure competitive positions for themselves and their countries when they grow up. Some issues are well known and have been of concern for several decades, whereas others are new.[13]

CONCLUDING REMARKS

Leaders of successful enterprises consistently search for ways to improve their organizations' performance. They wish to secure sustained

viability and success. Frequent disappointments with past management initiatives have motivated managers to obtain new understandings of the underlying but complex mechanisms – such as knowledge – that govern enterprise effectiveness and present-day markets.

KM, we believe, is far from the narrow management initiatives, or 'fads' such as TQM, BPR, downsizing, etc.[14] It is fundamentally different in both objective and scope. KM is broad, multi-dimensional and covers most aspects of the enterprise's activities. In contrast, fads have gained popularity by simplifying the problem setting and scope. Whereas simplicity has been their attractiveness, it is also their weakness.

To be competitive and successful over the long haul, experience shows that enterprises must create and sustain a balanced intellectual capital portfolio. They need to set broad priorities and integrate the goals of managing intellectual capital and the corresponding effective knowledge processes (EKPs). That requires systematic KM. It particularly is important to set priorities for which intellectual capital elements must be strengthened and which EKPs need to be undertaken to support the overall goals.[15]

Dependence on human intellectual functions in working life will change over future decades. This change is driven by the continued worldwide competitive forces with their increased reliance on personal and embedded knowledge. Success becomes a function of the quality of knowledge content available to create and deliver acceptable products and services, often tailored to individual customer's specific needs.

New workplace environments, advanced IT and organizational arrangements to promote collaboration, creativity, and performance improvements will create other changes – often in ways that we have not yet envisioned.[16] If we are fortunate, all these changes will provide greater personal leverage and lead to work lives that are more fulfilling.

With knowledge being a major driving force behind the 'economics of ideas' and hence behind new, resource-independent areas of growth, we can expect that emphasis on knowledge creation, development, organization, and leveraging will continue to be of prime focus for a long time. As a result, we should also find that continually improved and well applied knowledge will be the fuel to improve quality of life for the world-at-large. ❑

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[1] Reported by Wiig (1994) in Chapter 2 "Executive Perspectives on the Importance of Knowledge" (pp. 37-61).

[2] This approach was pursued by Digital Equipment Corporation in the early 1980s and in 1990 by the Initiative for Managing Knowledge Assets (IMKA). It recently has been pursued by many institutions, particularly those that focus on KM from the information technology perspective.

[3] Intellectual Capital Management (ICM) is led by Skandia, the Swedish insurance company and explained in the supplements to their 1994 and 1995 annual reports. (The approach may also be explored at http://www.skandia.se).

[4] Comprehensive KM has been described by Wiig (1994) and in greater detail by Wiig (1993) and Wiig (1995).

[5] Adapted with permission from Wiig (1995).

[6] Romer (1992) and Kelly (1996). See also the Internet World Wide Web: http://www.wired.com/4.06/romer/.

[7] Treacy and Wiersema (1993).

[8] See Gernot Böhme and Nico Stehr (1986), Harlan Cleveland (1985), Peter Drucker (1989) *The New Realities*, and Peter Drucker (1993) for further discussions of the knowledge society .

[9] See for example Simon (1976), Janis (1989), and Janis and Mann (1977).

[10] This listing reflects the author's limited insights and is clearly incomplete. Some additions have been provided by Amidon (1996), Demarest (1996) and Wilson (1996).

[11] Brooking (1996), Quinn *et al* (1996), Sveiby (1996).

[12] Amidon (1997), Spek and Spijkervet (1997), Stewart (1997).

[13] See for example Kenneth Boulding's lecture "The Image: Knowledge in Life and Society," (Boulding, 1956).

[14] Hilmer and Donaldson (1996).

[15] Amidon (1997), Brooking (1996), Spek and Spijkervet (1997), Stewart (1997), Sveiby (1996).

[16] Kao (1996).

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